Math Inquirers

Extension Workbook

with answers

Level 3

Self-study | Ages 8-10



My name _____

About this book

Math Inquirers Extension Workbook, Level 3
Self-study practice with answers.

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- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

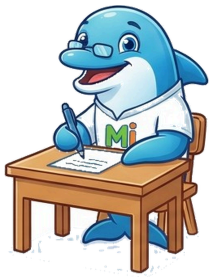
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	**Number sleuths**	6-7	8-9	10 and 12	11	☐
2	**Cryptarithm crackers**	Not in this proof edition				☐
3	**Times table treasure**	Not in this proof edition				☐
4	**Model makers (bar models)**	Not in this proof edition				☐
5	**Remainder riddles**	Not in this proof edition				☐
6	**Fraction walls**	Not in this proof edition				☐
7	**Area architects**	Not in this proof edition				☐
8	**Angle and line detectives**	Not in this proof edition				☐
9	**Measure and money masters**	Not in this proof edition				☐
10	**Puzzle day**	Not in this proof edition				☐
11	**Catch-up week**	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

Learn it: Numbers to 10 000

My goal. I can read, build and order 4-digit numbers. I can use clues to find a mystery number.

A 4-digit number has thousands, hundreds, tens and ones. The place decides what a digit is worth. A 0 can never be the first digit.

Worked example

Use the cards 7, 0, 4 and 9 once each. Make the greatest and the smallest 4-digit number.



1. **Greatest: put the greatest digit in the thousands place. Then the next greatest. This gives 9740.**
2. **Smallest: the first digit cannot be 0. Use the smallest digit that is not 0: the 4.**
3. **Put the other digits in order after it, smallest first: 0, 7, 9. This gives 4079.**

Answer: Greatest 9740. Smallest 4079.

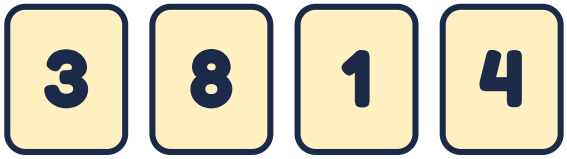
How to make a number from cards

1. Decide the thousands digit first.
2. Odd or even? Choose the ones digit first.
3. Making a list? Fix the first digit, then list the rest from smallest.

Your turn

Answer at the back of the book.

Try **Make the greatest even number with the cards 3, 8, 1 and 4. Use each card once.** ☐



Start here

Do these first. Take your time.

1 **Look at the place value chart.**

Th	H	T	O
4	0	7	2

(a) Write the number. ☐

(b) The digit 7 is worth ☐.

2 **Write each number in digits.**

(a) 6 thousands, 5 hundreds and 9 ones ☐

(b) 2 thousands and 15 tens ☐

(c) 5 thousands, 12 hundreds and 3 ones ☐

3 **Fill in the blanks.**

(a) 1000 more than 6457 is ☐.

(b) 100 less than 3050 is ☐.

(c) 10 more than 8995 is ☐.

(d) 1 less than 7000 is ☐.

4 **Arrange the numbers in order. Begin with the smallest.**
5830, 5083, 5380, 5038

_____

5 **A shop has 3 boxes of 1000 pins, 14 packs of 100 pins and 6 loose pins. How many pins are there in all?** ☐

Start here

Keep going. Check your work.

6 **Use the cards once each in every number.**



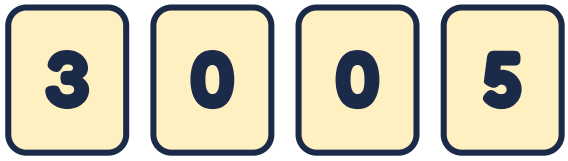
(a) The greatest number ☐

(b) The smallest number ☐

(c) The greatest odd number ☐

(d) The smallest odd number ☐

7 **List every 4-digit number that can be made with the cards 3, 0, 0 and 5. Begin with the smallest.**



_____

8 **Write the next 2 numbers in each pattern.**

(a) 2300, 2600, 2900, ☐, ☐

(b) 9000, 8750, 8500, ☐, ☐

(c) 3, 4, 6, 9, 13, ☐, ☐

9 **A pattern starts at 4000 and goes up by 500 each time.**

(a) Is 6800 in the pattern? Yes No

(b) The first number above 6800 in the pattern is ☐.

Start here

Nearly there. You can do it.

10 **Look at the number line.**



(a) Arrow A points to ☐.

(b) Arrow B points to ☐.

(c) Arrow C points to ☐.

11 **I am a 4-digit number. My thousands digit is 6, my hundreds digit is 3 less than that, and my tens digit is 0. My ones digit is the greatest odd digit.**

What number am I? ☐

12 **Ravi says the greatest odd number from the cards 9, 4, 6 and 2 is 9642.**

9	4	6	2

(a) Is he right? Yes No

(b) The greatest odd number from the cards is ☐.

13 **Look at the number 5737.**

(a) The first 7 is worth ☐.

(b) The second 7 is worth ☐.

(c) The first 7 is worth ☐ more than the second 7.

Think harder

Hints are upside down. Try first!

1 **The digits of a 4-digit number are all different and add up to 8. The number is even.**

(a) The greatest it can be is ☐.

(b) The smallest it can be is ☐.

Sets of digits: _____

Numbers I tried: _____

_____

_____

Hint. Find the sets of 4 different digits that add up to 8 first.

2 **Ali writes every number from 1 to 200. How many times does he write the digit 7?** ☐

1 to 99: _____

100 to 200: _____

Total: _____

_____

Hint. Count 1 to 99 first. Then count 100 to 200.

3 **Mei uses only the digits 1 and 2 to make 4-digit numbers. A digit can be used more than once. She writes them in a list, smallest first. Which number is 10th in her list?** ☐

My list: _____

_____

_____

Hint. Begin with 1111. Change the ones digit first. Keep a list.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 3: Counting digit sums.**
Count the 3-digit numbers with a digit sum of 4. List them by first digit.

(a) Digit sum 4: ☐ numbers.

(b) Use the pattern. Digit sum 5: ☐ numbers.

First digit 4: _____

First digit 3: _____

Hint. Fix the first digit. See how the counts grow: 1, 2, 3, …

2 **Beyond Level 3: Growing jumps.**
Look at the jumps between the numbers. Write the next 2 numbers.

5, 8, 14, 23, 35, ☐, ☐

Jumps: _____

Hint. Write the jumps between the numbers. What do you see?

3 **Make up your own number riddle. Choose a secret number with 4 digits, then write 3 clues. Test your clues: exactly ONE number must fit. Swap riddles with a grown-up.**

Clue 1: _____

Clue 2: _____

Clue 3: _____

Hint. Pick the number first. Then write clues about its digits.

4 **Find the secret number. It is greater than 3000 and less than 4000. Its digits are all different and add up to 15. The tens digit is 0, and the hundreds digit is double the ones digit.** ☐

Hint. Try ones digit 1, 2, 3 and 4. Check each sum.

✔ Check yourself

No hints. Try each one on your own.

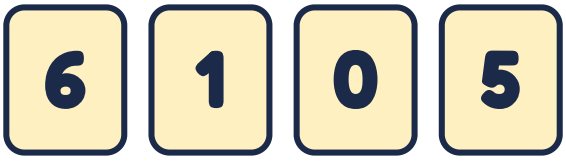
1 In 7258, the digit 2 is worth ☐.

Working: _____

2 Write the next 2 numbers. 6500, 6000, 5500, ☐, ☐

Working: _____

3 Make the smallest 4-digit number from the cards 6, 1, 0 and 5. Use each card once. ☐



4 Ali has 5 boxes of 1000 beads, 12 bags of 100 beads and 8 loose beads. How many beads does he have? ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Number sleuths	☐	☐	☐	☐	/ 4	
2	Cryptarithm crackers	☐	☐	☐	☐	/ 4	
3	Times table treasure	☐	☐	☐	☐	/ 4	
4	Model makers (bar models)	☐	☐	☐	☐	/ 4	
5	Remainder riddles	☐	☐	☐	☐	/ 4	
6	Fraction walls	☐	☐	☐	☐	/ 4	
7	Area architects	☐	☐	☐	☐	/ 4	
8	Angle and line detectives	☐	☐	☐	☐	/ 4	
9	Measure and money masters	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 3.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 1 · Number sleuths

Your turn
page 6

Answer: 8314
Why: An even number ends in 8 or 4. With 4 last, the other cards in order give 8314. With 8 last, the best is 4318. So 8314 is greater.

START HERE

1
page 7

Answer: (a) 4072 (b) 70
Why: Read the places from the left: 4 thousands, 0 hundreds, 7 tens and 2 ones make 4072. The 7 is in the tens place, so it is worth 70.

2
page 7

Answer: (a) 6509 (b) 2150 (c) 6203
Why: (a) 6000 + 500 + 9 = 6509. (b) 15 tens is 150, so 2000 + 150 = 2150. (c) 12 hundreds is 1200, so 5000 + 1200 + 3 = 6203.

3
page 7

Answer: (a) 7457 (b) 2950 (c) 9005 (d) 6999
Why: (a) Only the thousands digit changes: 7457. (b) 3050 - 100 = 2950. (c) 8995 + 10 = 9005, and the 9 hundreds carry. (d) 7000 - 1 = 6999.

4
page 7

Answer: 5038, 5083, 5380, 5830
Why: All four start with 5 thousands. Compare the hundreds next: 0, 0, 3, 8. Then compare the tens of 5038 and 5083: 3 is less than 8.

5
page 7

Answer: 4406
Why: 3 boxes are 3000 pins. 14 packs are 1400 pins. 3000 + 1400 + 6 = 4406.

6
page 8

Answer: (a) 8752 (b) 2578 (c) 8725 (d) 2587
Why: (a) Greatest digits first: 8752. (b) Smallest digits first, no 0 here: 2578. (c) The ones digit must be 7 or 5. With 5 last, 8725 beats 8527. (d) With 7 last, 2587 beats 2785.

7
page 8

Answer: 3005, 3050, 3500, 5003, 5030, 5300
Why: The first digit is 3 or 5, never 0. After 3, the 5 can go in the hundreds, tens or ones place: 3005, 3050, 3500. After 5, the same way: 5003, 5030, 5300. That is 6 numbers.

8
page 8

Answer: (a) 3200, 3500 (add 300) (b) 8250, 8000 (subtract 250) (c) 18, 24
Why: (a) Add 300 each time. (b) Subtract 250 each time. (c) The jumps are 1, 2, 3, 4, so the next jumps are 5 and 6: 13 + 5 = 18 and 18 + 6 = 24.

9
page 8

Answer: (a) No (b) 7000
Why: The pattern goes 4000, 4500, 5000, 5500, 6000, 6500, 7000. It jumps over 6800 and lands on 7000.

10
page 9

Answer: (a) 6150 (b) 6450 (c) 6350
Why: Each small mark is 50. A is halfway from 6100 to 6200: 6150. B is halfway from 6400 to 6500: 6450. C is halfway from 6300 to 6400: 6350.

11
page 9

Answer: 6309
Why: The thousands digit is 6. The hundreds digit is 6 - 3 = 3. The tens digit is 0. The greatest odd digit is 9. The number is 6309.

12
page 9

Answer: (a) No **(b)** 6429
Why: 9642 ends in 2, so it is even. The only odd card is 9, so 9 must be in the ones place. The other cards in order are 6, 4, 2. The number is 6429.

13
page 9

Answer: (a) 700 **(b)** 7 **(c)** 693
Why: The first 7 is in the hundreds place: 700. The second 7 is in the ones place: 7. 700 - 7 = 693.

THINK HARDER

1
page 10

Answer: (a) 5210 **(b)** 1034
Why: Only two sets of 4 different digits add up to 8: 0, 1, 2, 5 and 0, 1, 3, 4. Even numbers need an even last digit. The greatest even number is 5210. The smallest is 1034.

2
page 10

Answer: 40
Why: From 1 to 99: the digit 7 is in the ones place 10 times (7, 17, ... 97) and in the tens place 10 times (70 to 79). That is 20. From 100 to 200 it is 20 again. Total 40.

3
page 10

Answer: 2112
Why: The list goes 1111, 1112, 1121, 1122, 1211, 1212, 1221, 1222. That is 8 numbers, all starting with 1. Next are 2111 (9th) and 2112 (10th).

GO FURTHER

1
page 11

Answer: (a) 10 **(b)** 15
Why: Sum 4: first digit 4 gives 1 number (400), 3 gives 2 (310, 301), 2 gives 3, 1 gives 4. That is 1 + 2 + 3 + 4 = 10. Sum 5 goes up to 5 numbers for first digit 1: 1 + 2 + 3 + 4 + 5 = 15.

2
page 11

Answer: 50, 68
Why: The jumps are 3, 6, 9, 12. Each jump is 3 more than the one before. The next jumps are 15 and 18: 35 + 15 = 50 and 50 + 18 = 68.

3
page 11

Answer: Answers vary. Example: I am between 7000 and 8000. My hundreds digit is double my tens digit. My ones digit is 1 more than my tens digit. My digits add up to 16. The only number is 7423.
Why: Check that every clue is true for your secret number. Then list the numbers that fit clue 1, and cross out those that fail clue 2, and so on. In the example, the tens digit t gives 7 + 2t + t + (t + 1) = 16, so t = 2 and the number is 7423.

4
page 11

Answer: 3804
Why: The thousands digit is 3 and the tens digit is 0. If the ones digit is o, the hundreds digit is 2o, so 3 + 2o + 0 + o = 15 and o = 4. The hundreds digit is 8. The number is 3804.

CHECK YOURSELF

1
page 12

Answer: 200
Why: The 2 is in the hundreds place, so it is worth 200.

2
page 12

Answer: 5000, 4500
Why: Subtract 500 each time: 5500 - 500 = 5000 and 5000 - 500 = 4500.

3
page 12

Answer: 1056
Why: The first digit cannot be 0, so use 1. Then the rest in order: 0, 5, 6. The number is 1056.

4
page 12

Answer: 6208
Why: 5 boxes are 5000. 12 bags are 1200. 5000 + 1200 + 8 = 6208.